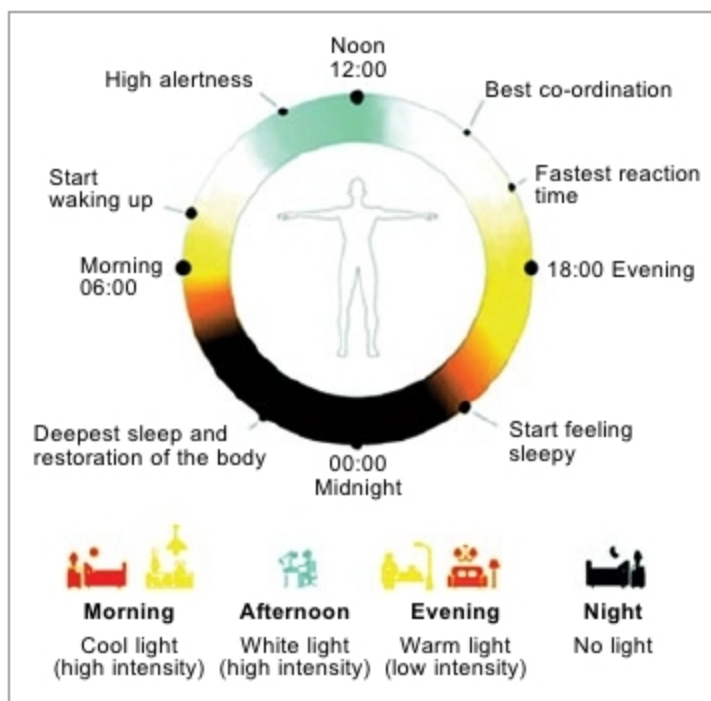




Human-centric lighting (Credit: www.ee.co.za)

Sumit Joshi, vice chairman and managing director, Philips Lighting India, shares, “Connected lighting offers various insights about a building, by integrating seamlessly with the IT systems in the building, demonstrating how dynamic buildings can become smarter and more efficient than ever before.

“Connected luminaires can be fitted with sensors that can enable managers to track occupancy patterns, changes in temperature, light levels and more, while employees can personalise the lighting around their desks. The system allows maximum visibility and better control. It allows offices to reduce energy consumption based on building usage and occupancy. Because fixtures receive both data and power over a single Ethernet connection [also called Power over Ethernet (PoE)], smart lighting also eliminates the need for costly electrical wiring during construction or renovation.”

Integrating workspace design with human-centric lighting

Different criteria like glare control, contrast sensitivity, illuminance and uniformity define the quality of light. Explaining the technological elements, Ripu Daman Sharma, country sales manager, Lutron electronics, says, “Wireless lighting utilises the intelligence of software technology and sensors to make people’s lives simpler and more convenient. It is designed for energy efficiency. The technology

allows LEDs to sense where the light is needed and accordingly sets different colours and intensity of light. Simply put, smart or wireless lighting is associated with a system that provides centralised lighting control.”

“Philips Lighting is allowing Internet-connected lighting to be paired with PoE technology. This enables routing of low-voltage electricity over Ethernet cables of a smart building’s IT network to power LED luminaires. Ethernet cables help carry data to and from the lights, allowing building occupants and

operators to intelligently control lights from Internet-connected computers, smartphones and tablet devices, either onsite or remotely,” shares Joshi.

According to a research done by Philips, human-centric lighting is focused on indoor illumination. It is related to mesopic vision, where the eyes are sensitive to dimmed levels.

Light, especially blue wavelengths, during the evening hampers the sleep process. For example, the huge billboards on buildings play a negative role in the sleep-wake cycle.

The challenge today is to create energy-efficient environments while energising people. LEDs and connected technology take care of this. Economic value of human-centric lighting comes in the form of increasing employee productivity.

Controlling the future

The outcome of bringing about this turnaround in lighting is to be able to control our energy consumption and blend our artificial environment with the natural one, as much as possible.

The future is completely in our control. Joshi concludes, “Lighting is an integral part of any building’s infrastructure and, hence, can play an important role in creating sustainable structures. With the introduction of energy-efficient LEDs, power consumption of lighting has been reduced significantly. This can be further reduced with connected LED lighting, which can be monitored and controlled remotely by building administrators.”

“By tracking occupancy patterns, changes in temperature and light levels, the system allows maximum visibility and control. It also allows building administrators to manage energy consumption based on usage and occupancy. When not in use, lights can be dimmed, significantly reducing overall energy consumption.

“Connected lighting can significantly reduce the overall carbon footprint. Typically, lighting is responsible for 40 per cent of a building’s electricity use. Internet of Things (IoT)-enabled lighting could dramatically reduce this energy consumption, and lead to about 80 per cent savings.”

—Nidhi Arora, executive editor, EFY

Connected Manufacturing: Auto-Parts Manufacturer Leverages The Industrial IoT

➤ With the adoption of the IoT, manufacturers can gain enhanced transparency, traceability and transformation

Connected technology for factories has the potential to bring vast business transformations. By driving infrastructure optimisation and predictive maintenance, it induces substantial cost reduction and productivity improvement. Abhijit Mhetre, vice president - marketing, Altizon Systems, says, “With the adoption of the IoT, manufacturers can gain enhanced transparency, traceability and transformation. The ability to segment, analyse and correlate machine